201. We have a microphone in the lab with a sensitivity of -50dBV/Pa. What will be the pressure on this microphone when it generates a level of -60dBV?

202. We have two microphones, m_1 and m_2, located at the same position, capturing the same source, s_1. The ambient noise ($P_{AMBIENCE}$) corresponds to 30dB$_{SPL}$. The first microphone (m_1) outputs -40dBV when the source generates its maximum pressure (P_{MAX}). The output of both microphones is summed through a mixing console at unity gain for each channel (0dB). The result of this addition generates -36dBV for P_{MAX}.

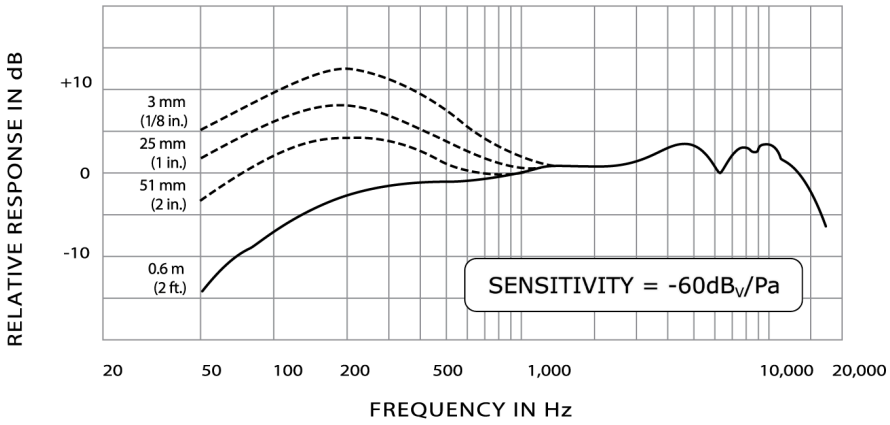
a. If the sensitivity of m_2 is -51dBV/Pa, what is the maximum pressure captured by the microphones? What is the sensitivity of m_1?

b. If m_1 and m_2 were exactly the same microphone, what would be their sensitivity if the output of the microphone is -36dBV for P_{MAX}?

c. Considering the different microphone's sensitivity case study, what would be the dynamic range at the master output of the mixing console for P_{MAX}? (Note: use uncorrelated addition as the ambient noise is a random signal).

d. Compare the dynamic range obtained in the mixing console with the acoustic dynamic range, using 20 $\log_{10}(P_{MAX} / P_{AMBIENCE})$. Are they the same? Why?

203. We want to make a live recording of a pop-rock singer with an unknown microphone. The only information about the microphone is the following chart:

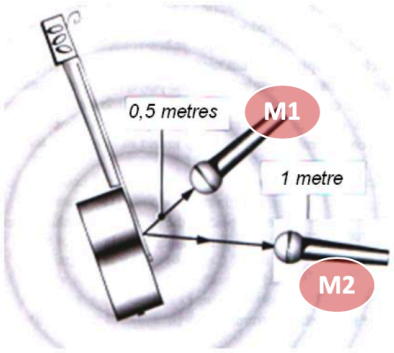


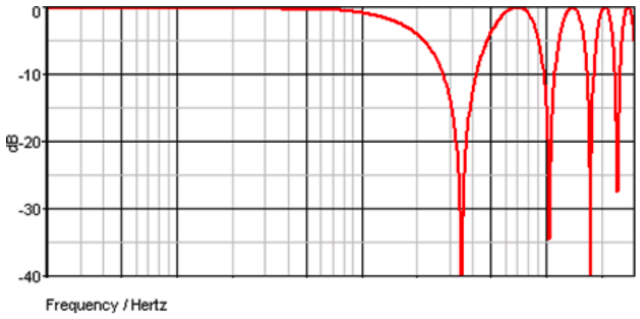
a. Determine all that is deductible in terms of: type of transduction, frequency response, directivity pattern, sensitivity, etc...

b.Whether the nominal sensitivity of the microphone corresponds to -60dBV/Pa and the singer is 3mm far from the capsule, how many Volts will we obtain at 100Hz for an incident pressure of 3Pa?

c.Do you think that the value obtained in the previous section (b) may depend on the way the singer holds the microphone? Explain why.

204. We have two microphones, m_1 and m_2, respectively located at 0.5m and 1m from a classical guitar.





The guitar is located in a very dry room (almost anechoic), with a background noise level of $32dB_{SPL}$. The m_1 outputs -60dBV when the instrument generates its maximum pressure (P_{MAX}). The outputs of m_1 and m_2 are summed in a mixing console at unity gain for each channel (0dB). The result of this addition generates -57dBV at P_{MAX}.

a. If the sensitivity of m_2 corresponds to -50 dBV/Pa, what would be the maximum pressure of the guitar in dB_{SPL} at m_1?

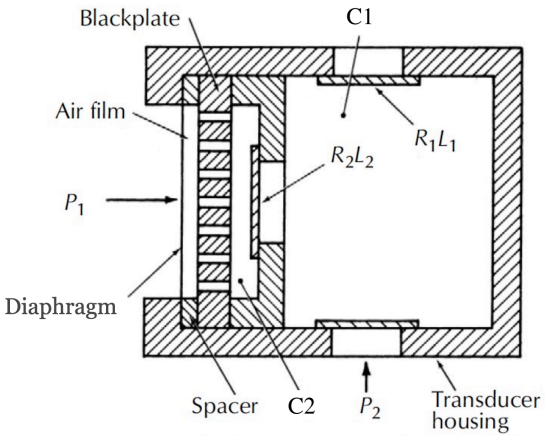
b. What would be the m_1 sensitivity?

c. Considering that the background noise in a room is random (non coherent), what would be the dynamic range at the master output of the mixer?

d. What would be the acoustic dynamic range in the room for the m_1 placement? Explain qualitatively the difference among the two dynamic range values.

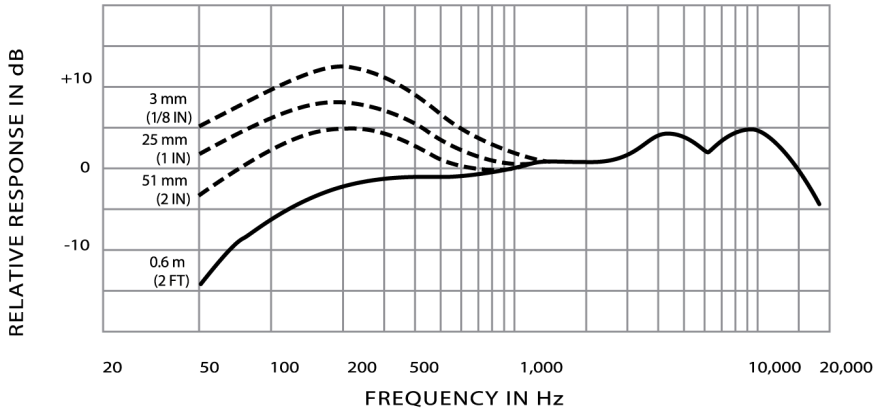
e. Which frequency phenomenon do we consider when the two signals are summed? Relate it to the figure on the right and give values for the first constructive and destructive interferences.

205. Obtain the equivalent circuit of the following microphone capsule for both, the impedance analogy and the mobility analogy.

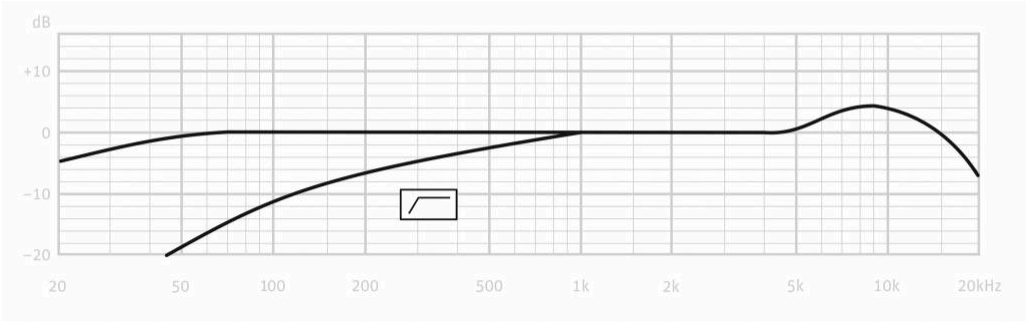


206. We have two perfectly aligned microphones (M1 and M2) placed at 0,8m from a loudspeaker that is radiating a sine wave of 1KHz. The next graphs show the frequency responses and specifications of the two microphones:

M1: Supercardioid. Sensitivity = -51.5 dBV/Pa



M2: Omnidirectional response. Sensitivity = 20 mV/Pa



a. If the SPL at the position of the microphones is 91,94 dB, calculate the equivalent pressure.
b. Calculate the total level in dBu that we could measure in a mono bus of a mixing console if we add the contributions of the two microphones with all the amplification stages at 0dB.
c. Calculate the gain in dB needed for M1 to match the level of M2 in the previously described position.

Now we change the test signal to a sine wave of 200Hz, and with a sonometer we measure 80 dBSPL at the same position of the microphones (0,8m).

d. Calculate the voltage in dBV obtained in M2 at 0,8m if we activate its Low Cut filter.
e. Calculate the voltage in dBV obtained in M1 if we place the microphone at 2,5 cm from the loudspeaker.
f. Calculate the first frequency canceled in the final combined response if we add the signals of the two microphones in a mono bus of the mixing console. (c=344 m/s).